Behavioral Analytics of Subscription Upgrades and Cross-Selling: A Data-Driven Strategic Approach

Abstract

Purpose: This research aims to use behavioral analytics and machine learning to determine how customers are likely to upgrade their subscription and what cross-selling opportunities to consider in an online subscription product and it aims at allowing companies to make informed and data-driven strategic choices in order to improve customer engagement and lifetime value.

Design/methodology/approach: This study relies on secondary data sources. An organized analytical system was applied, which started with data preprocessing, missing values, and coding. Behavioral rules and consumption patterns were used to construct target variables of upgrade and cross-sell prediction. To test the predictive ability of logistic regression, random forest, and XGBoost machine learning models were developed and compared based on their predictive capacity. The classification measures that were used to evaluate model performance were based on hold-out testing.

Findings: Behavioral engagement features strongly predict subscription upgrade decisions, with stable performance across models. Cross-sell behavior is more complex and less predictable, reflecting distinct customer expansion dynamics.

Research limitations/implications: The limited scope of one-platform dataset and the rule-based method of defining the label of cross-sell restrict the study and present the possibility of introducing subjective bias. Wider data can make it generalize better.

Practical implications: The predictive framework is actionable, giving subscription companies insight into how to upgrade per user individually, improve cross-selling tactics, and improve efficient customer retention programs.

Originality/value: The work is also a novel contribution to the field of subscription service analytics through the combination of behavioral feature engineering and predictive modeling as

it combines to tackle the issue of upgrade prediction and cross-selling at the same time to provide a single and operationally applicable decision-support mechanism to digital service providers.

Keywords: Consumer Behavior Analytics, Subscription Upgrade, Cross-Selling, Predictive Modeling, Strategic Decision-Making, Customer Lifetime Value, Behavioral Insights

1. Introduction

The accelerated international growth of the subscription-based business model has altered the way organisations collect income, create lasting customer connections, and create tailored online experiences (Lindström et al., 2024). Subscriptions focus on the constant interaction, retention, and expansion, which involve subscription upgrades and cross-selling, in place of a single transaction (Janzer, 2020). As the global subscription economy is expected to achieve total growth of about 67% over five years and approximately $1.2 trillion in 2030 (Juniper Research, 2025), customer behavior has become a key strategy ability of digital service providers (Lee & Lee, 2020).

Behavioral analytics - the behavior of users, consumption, preferences, and indicators of engagement - offers understanding of the manner and reason why a customer interacts with digital services (Theodorakopoulos & Theodoropoulou, 2024). Previous studies indicate that customer expansion strategies like upselling and cross-selling have a significant impact on profitability, with these two approaches likely to drive half of all revenues in an increase of digital subscriptions companies (Pandya & Dholakia, 2021). Nevertheless, even with the presence of rich behavioral data, intuition-based or rule-based targeting remains used by numerous organizations (Salas et al., 2010), which could fail to capture latent behavioral signals that can be more readily recognized by machine learning models.

There is a research gap in the fact that very little literature is combined to offer behavioral feature engineering and predictive analytics to identify all opportunities of subscription upgrades and cross-selling simultaneously. The available literature is mostly churn prediction or upgrade prediction, but only a small number of studies present a unified prediction model that has been constructed directly based on behavioral patterns (Alboukaey et al., 2020). Also, contemporary subscription ecosystems have more behavioral variables to capture, in addition to the focus on demographic or transactional variables, as observed in the literature.

In order to fill this gap, the current research constructs a holistic information-based strategic framework that incorporates behavioral analytics and machine learning methods. The methodology involves systematic preprocessing of data, treatment of missing values, encoding customer attributes, building rule generated target variables to upgrade and cross-sell probability, and supervised learning algorithms which are Logistic Regression, Random Forest, and

XGBoost. Through these models, it is possible to identify customers who are most likely to upgrade into higher levels of subscription or embrace other services. Comparison of performance of the model in terms of classification metrics helps to identify the behavioral indicators with the highest predictive power (De Cnudde et al., 2020).

The results show that the ensemble learning models are more accurate and that they behave more in terms of the depth of the interactions of behaviors. The length of views, regularity of payments, use of device, and intensity of consumption of content become important behavioral indicators of upgrade and cross-sell intentions. The outputs provide real-world use to subscription companies that need to develop a customized recommendation, customer lifetime value, and effective retention and growth strategies.

In sum, this paper can contribute to the current expanding field of customer analytics because it presents a whole, behavior-based predictive model that can be used to optimize both upgrade and cross-sell requests. Through machine learning and understanding of behavior, organizations have the opportunity to shift their reactive marketing strategies to proactive and data-driven customer expansion strategies (Van Chau & He, 2024; Ramya et al., 2024).

2. Literature Review

2.1 Evolution of Subscription-Based Business Models

The emergence of the subscription-based business models has brought a significant change in the way organizations acquire, maintain, and develop their customers. The subscription economy of continuous engagement and recurring revenue is growing at an extremely fast pace that has been witnessed in the entertainment industry, telecommunications industry, e-commerce, digital services among others (Verma et al., 2025). Expanding the global digital presence and the desire of customers to pay to access it constantly rather than owning are all positive factors to develop this model. This change underscores the necessity of the companies to leave the old marketing metrics and implement customer lifetime value (CLV) (Ali & Shabn, 2024) and customer expansion strategies like upselling and cross-selling, which are regarded as the key foundations of profitability in subscription ecosystems (Seidenstricker, 2024).

2.2 Subscription Service Behavioral Analytics

Behavioral analytics is dedicated to the interaction of the customers with a service; what content they see, how frequently they use particular features, the view time, their device preferences and the payment frequency (G. Martín et al., 2021). Previous studies underline the fact that behavioral variables tend to be a better forecaster of customer intent in online settings than demographic or transactional variables (Ranaweera et al., 2005; Abbasi et al., 2015). As an example, the higher the frequency of use, the more content has been consumed and the more time one has been engaged, the higher the upgrade or retention chances (Kim et al., 2013). This is because extracting behavioral insights based on customer behavior can enable a firm to comprehend the motivational factors of an individual and develop offering strategies that are more applicable (Cao, 2010; Theodorakopoulos & Theodoropoulou, 2024).

2.3 Customer Decision Making Predictive Modeling

Machine learning methods have become popular in forecasting customer responses like churn, retention, upgrade and use of the new product (Adekunle et al., 2023; Khodabandehlou & Zivari Rahman, 2017). Conventional statistical control models such as the use of logistic regression have been extensively used because they are interpretable but current researches show that other

advanced algorithms such as the Random Forest, the Gradient Boosting Machines and XGBoost are more accurate in detecting complex behavioral patterns (Kavzoglu & Teke, 2022). Such models predict non-linear dynamics and latent relations among two or more behavioral attributes, which is why they are good in the context of subscription and digital analytics.

Various studies have highlighted how ensemble models are effective in predicting decisions of customers (Coussement & De Bock, 2013). proved that tree-based ensemble models outperform other models in churn prediction, whereas Saravi et al. (2022) indicated that the models that are based on statistical and machine learning predictors lead to better accuracy in decision-making. These results justify the applicability of ensemble learning methods to predict the behavior of upgrade and cross-selling, in which indicators of behavior can be interacting in several complex ways.

2.4 Cross-Selling and Upselling Strategies

The essential elements of the customer expansion management are the cross-selling and upselling (Vatavwala et al., 2023). Cross-selling will help promote buying additional services that will complement each other, whereas upselling will promote the purchase of a more advanced subscription (Solberg Söilen, 2024). Customer relationship management (CRM) studies have emphasized that frequently the income on customer extension can be more than the income on the first acquisition (Verhoef, 2003). To digital service providers, these strategies are a major contributor to average revenue per user, as well as long-term profitability.

Research demonstrates that data based cross selling techniques may enhance the conversion rates and customer satisfaction as they make the recommendations context based (Ghoshal et al., 2021; Saleem et al., 2019). Behavioral predictors e.g. content engagement, device use, viewing, and method of payment are critical in influencing the receptiveness of a customer to cross-selling propositions. Furthermore, predictive analytics-based personalization proved to be more responsive to customers as they are offered with the help of previous behavior (Madanchian, 2024).

3. Research gap and conceptual framework

3.1 Conceptual Contribution and Research Gap

Digital services that are delivered on a subscription basis are becoming more dependent on data-driven decision-making in order to grow revenues by retaining and acquiring customers. The current literature has already examined customer analytics in terms of churn prediction, customer lifetime value (CLV) prediction, and recommendation systems. Nevertheless, the analytical emphasis of explicit behavioral modeling of subscription upgrades and cross-selling decision as different predictive objectives is limited. The proposed study, which is a unified but differentiated modeling framework of upgrade and cross-sell prediction by behavioral engagement features, addresses this gap.

3.1.1 Positioning Within Existing Literature

Prior research on customer analytics can be broadly classified into four dominant streams:

1. **Churn Prediction Models:** The literature on churn prediction is mainly dedicated to finding out potential customers that will quit using the services. These models are risk-focused models and concentrate on early warning of the disengagement signs like the decreasing number of usage or the decrease of the activity. Although the churn models can be used in retention, they fail to consider the customers who are active yet they might want to increase the spending in terms of upgrades or adoption of additional products. As a result, the churn-focused strategy provides little information on the prospects of proactive revenue growth.
2. **Recommendation Systems:** The recommendation systems are commonplace such that products or content are suggested using the historical preferences, the collaborative filtering or the similarity measures. Though these systems can be used to perform real-time personalization, they are not directly tailored to predict the possibility of converting such monetization events as subscription upgrades or cross-selling. They also focus more on matching preference as opposed to behavioral preparation to commit to financial engagements, and thus do not give them an appropriate role in strategic revenue forecasting.
3. **Customer Lifetime Value (CLV) Forecasting:** The CLV models are used to determine the monetary worth of consumers in the long term through the aggregation of projected revenues in the future. Although CLV offers a macro-financial view, it is not very specific in

determining the specific behavioral trigger that enables an upgrade or cross-sell decision to be made. In addition, CLV models usually underline the assumption of constant behavioral patterns, and do not differentiate the various revenue-generating actions.

4. **Upgrade and Cross-Sell Behavior-Modeling:** In comparison with the areas mentioned above, there is a lack of behavioral modeling on the issue of upgrade and cross-sell decisions. The available literature tends to discuss upgrades and cross-selling as the byproducts or incorporate them in the framework of recommendation or CLV. There are not many studies that directly model these results as independent predictive behaviors based on patterns of behavioral engagement.

3.1.2 Identified Research Gaps

1. **Conceptual Gap:** Previous research fails to distinctly distinguish between two behaviors (churn prevention and revenue expansion). Upgrade and cross-sell choices are proactive and customer value-enhancing choices that should be conceptually different and not subsumed by retention or recommendation frameworks.
2. **Methodological Gap:** Few studies compare interpretable statistical models and state-of-the-art machine learning algorithms explicitly on the attribute of upgrade and cross-sell prediction based on the same space of behavioral features. This limits the knowledge of the suitability of models in various monetization purposes.
3. **Behavioral Gap:** Inadequate research has been done on the differentiated impacts of the intensity of engagement, content usage, and frequency of interaction on the results of upgrade and cross-sell outcomes.
4. **Practical Gap:** The available models present a few actionable marketing advice to marketing managers who want to implement specific upgrade and cross-sell campaigns, using predictive information.

3.1.3 Conceptual Contribution of the Study

- It also conceptually differentiates the upgrade and cross-sell decisions as two different customer expansion behaviors, which are related to each other.
- It presents a behavioral feature engineering model that reflects the intensity of use, the degree of engagement, and diversity of interaction.

- It empirically shows that upgrade behavior has stronger linear and explainable tendencies, and cross-sell behavior is more intricate and non-linear and has varying modeling anticipations.
- It compares and contrasts the models of Logistic Regression, Random Forest, and XGBoost using the same experimental design and provides the methodological clarity.

In this way, the research takes the literature on customer analytics beyond the churn and recommendation paradigms to a behavior-based monetization modeling paradigm.

3.2 Conceptual Framework

This research is based on the theory of customer engagement and relationship marketing. It hypothesizes that the customer engagement in behavioral terms has a direct impact on the monetization outcome in the platform that is based on subscription.

The behavioral engagement is made measurable using the indicators of content consumption, frequency of interaction, and intensity of platform use. These indicators can be used as explanatory variables that drive two dependent variables, i.e., subscription upgrades and cross-selling adoption.

Notably, the framework presupposes that both outcomes are the consequence of engagement, yet their mechanisms of decision-making are different. Increase in perceived value of the core service influences upgrade decision whereas exploratory behavior and complementary product interest influences cross-sell decision.

3.3 Research Objectives and Research Questions

The main aim of this study is to use behavioral analytics based on machine learning to find and forecast customer monetization opportunities in a digital service setting that utilizes subscriptions. To help accomplish this overall objective, the paper is informed by the following questions:

1. To investigate the behavioral and engagement patterns of customers and their impact on the subscription upgrade and cross-selling.
2. To assemble business-inspired target variables to determine the potential subscription upgrades and cross-selling prospects using behavioral analytics.

3. To create machine learning models (Logistic regression, random forest, and XGBoost) to predict upgrade readiness and probability of cross-selling.
4. In order to determine model performance in terms of classification metrics and ROC-AUC analysis to determine predictive performance.
5. To create data-driven insights that can be used in strategic decision-making to manage subscription and maximize revenues.

4. Methodology

4.1 Data Source and Preprocessing

The research has a structured customer-level subscription dataset that aims at capturing user demographics, subscription features, behavior and intensity of engagement and usage. The data contains variables of content consumption, frequency of interaction, tenure of accounts, and monetization-indicators, variables that are frequently used in subscription analytics and customer lifecycle modelling.

In the preprocessing, detection and removal of non-informative and identifier-based attributes (including unique customer IDs) were also done since these are not related to predictive learning and also produce noise. The missing values were evaluated on a feature-by-feature basis and dealt with by applying relevant statistical imputation techniques depending on the type and distribution of the variables. The imputation of numerical variables was done by the central tendency measures and the mode-based imputation of categorical variables was done where necessary.

Types of data were standardized to make sure that numerical features are properly interpreted by machine learning algorithms. Before modeling, exploratory data analysis (EDA) was performed to analyze the distributions of the variables, identify possible skewness, possible outliers, and realize the correlations between features of engagement and customer behavior. This measure made sure that the dataset was analytically sound and appropriate to do the downstream modeling work.

All preprocessing activities were done prior to model training to prevent data leaking, as well as the consistency of transformations between training and testing sets.

4.2 Construction of Intent Based Target Variable

First trials of modelling upgrade and cross-sell behaviour in terms of direct transactional or outcome-based measures were harshly class imbalanced whereby the positive classes had a very small percentage representation in the dataset. This imbalance greatly lowers the capability of a model to learn meaningful patterns and in most cases gives biased predictions towards the majority class.

To counteract this constraint, the study was based on intent based target variable construction strategy. The targets were specifications which aimed at capturing the customer readiness or propensity instead of modeling the outcome of confirmed purchases using the observed behavioral cues. It is a more realistic business practice model because predictive models are more commonly employed in identifying the high-potential customers to implement proactive marketing interventions than operate after the fact with the classification of outcomes.

Two binary target variables were created:

- **UpgradeTarget:** This is a term used to denote the customers with a high probability of moving to a higher level of subscription. The indicators that were used to derive this target were the intensity of content consumption, level of engagement on the platform, monetization behavior, and account maturity.
- **CrossSellTarget:** Refers to customers that exhibit behavioral interest that is competent to cross-sell other services or add-ons. Patterns of engagement breadth, interaction diversity, and content exploration were used to construct this target.

The intent-based algorithm dramatically enhanced the balance of classes and retained good business applicability so that models learnt behavioral patterns that are doable, rather than infrequent transactional phenomena.

4.3 Label Leakage Prevention

In order to ensure the rigor of the methodology, specific care was taken to avoid the leakage of labels, where predictors have direct or indirect information that is used to form the target variable.

The feature set used in the training of the model was restricted to all the variables that were directly related to the logical rules determining UpgradeTarget and CrossSellTarget. This made sure that there was no easy way of having models recreates the target definition, but instead they had to learn using independent behavioral and demographic cues.

Also, although CrossSellTarget was incorporated into the model of predicting the upgrade as an explanatory variable and thus that would capture the realistic customer progression dynamics, no future or post-outcome data was added. Every feature employed was the one that was current and

could be observed at the moment of prediction, maintaining the temporal consistency, as well as the causal validity.

4.4 Feature engineering and transformation

The feature engineering had been intentionally restrained and interpretable to prevent overfitting and inflated artificial performance. Some of these transformations were chosen to enhance signal-to-noise ratio by capturing relative engagement intensity and usage efficiency in composite and ratio based indicators.

The numerical amounts were normalized with z-score to ensure equal scale especially with linear model like Logistic Regression which is sensitive to feature magnitude. The transformation also enhanced the stability of convergence and model coefficients interpretability.

There were no synthetic resampling methods like SMOTE used, because the refined target construction plan could obtain a satisfactory balance of the classes. This allowed model performance to capture actual data trends and not man-made observations.

4.5 Train-Test Split and Validation Strategy

An 80:20 split was used to split the dataset into training and testing sets. To ensure fair and unbiased assessment, stratification on the basis of the target variable was used so that the proportions of classes in both of the subsets were maintained.

In order to evaluate further model stability and generalization performance stratified 5-fold cross-validation was used in training. By doing so, it was possible to average performance metrics using multiple data partitions rather than being reliant upon a single train-test split and enhance the quality of model comparison.

4.6 Model Development

Three supervised machine learning classification algorithms were written and tested in both upgrade and cross-sell prediction tasks:

- Logistic Regression
- Random Forest
- Extreme Gradient Boosting (XGBoost)

An underlying model was Logistic Regression, as it can be interpreted and is probabilistic, not to mention its use is common in customer analytics. The tree-based ensemble was added to ensure that non-linearism of relationships and effects of interaction, which are usually involved in customer engagement behavior, are captured.

The imbalance between classes was resolved with algorithm-specific weighting approaches, including class weights in Logistic Regression, scale adjustments in ensemble models, and so on, so that the minority class predictions could be correctly punished in the training process.

4.7 Model Evaluation Metrics

The Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) was used in the evaluation of model performance. ROC-AUC was used because it is a threshold independent measure of how well a model can separate the classification of the classes especially when there is an imbalance among the classes.

Further measures of evaluation were calculated using precision, recall, F1-score, and confusion matrices as well to determine the quality of the classification using statistical viewpoint as well as business viewpoint. These indicators allowed to analyse the trade-offs between false positives and false negatives that play a pivotal role in marketing decision-making situations.

The threshold tuning was also conducted to ensure the maximization of the balance of recall-precision, which aligned predictions with the feasible campaign targets.

4.8 Model Selection Criteria

A combination of predictive performance, cross-validation folds consistency and interpretability were used to select the final model. To predict upgrades, the Logistic Regression showed a better ROC-AUC and a consistent performance, thus can be used in deployment where transparency and explainability are needed.

To model complex, non-linear patterns of engagement, tree-based models demonstrated a relatively better performance in terms of cross-sell prediction. These models were thus deemed to be more suitable in identifying cross selling opportunities that were instigated by various behavioral signals.

5. Results and Analysis

In this section, the empirical findings of the machine learning models that were designed to forecast the subscription upgrading and the cross-selling opportunities are presented. Three classification algorithms were compared, including Logistic Regression, Random forest and XGBoost with stratified cross-validation and independent test set. ROC-AUC, confusion matrices and conventional classification metrics were used to test model performance.

5.1 Performance Evaluation of Upgrade Prediction Models

5.1.1 Logistic Regression Results (Upgrade Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.8564**
Standard Deviation (ROC–AUC)	**0.0232**

Table 5.1 ROC-AUC Performance

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.85	0.94	0.89	719
Yes	0.74	0.50	0.60	244
Accuracy	—	—	**0.83**	963
Macro Average	0.79	0.72	0.75	963
Weighted Average	0.82	0.83	0.82	963

Table 5.2 Classification Report

Fig. 5.1 Confusion Matrix

Logistic Regression proved to be the best when compared to all considered predictive models of upgrades. The ROC-AUC mean of the model was 0.856 (±0.023) which is high quality discriminative power. As it can be seen in the confusion matrix, the model has identified a substantial percentage of the non-upgrade customers and is moderately sensitive to the upgrade-prone customers.

The classification report indicates the classification report has a precision of 0.74 and recall of 0.50 concerning the positive (upgrade) category. It represents a trade-off of finding real upgrade customers and false positives. The high accuracy of 83% also proves the strength and constancy of the model.

Interpretation:

The interaction patterns in upgrade behavior are relatively clear and consistent, and reasonably represented by a linear decision boundary. Logistic Regression is very appropriate in upgrade targeting due to their interpretability and high predictive power.

5.1.2. Random Forest Results (Upgrade Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.8296**
Standard Deviation (ROC–AUC)	**0.0278**

Table 5.3 ROC-AUC Performance

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.84	0.92	0.88	719
Yes	0.68	0.49	0.57	244
Accuracy	—	—	**0.81**	963
Macro Average	0.76	0.70	0.72	963
Weighted Average	0.80	0.81	0.80	963

Table 5.4 Classification Report

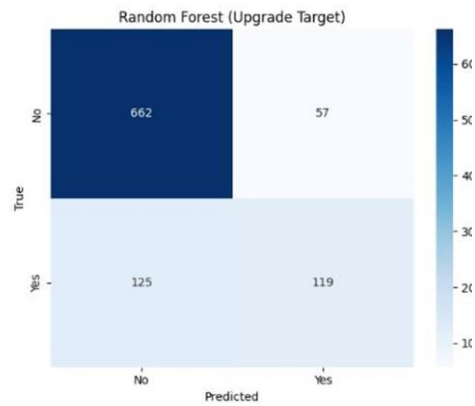

Fig. 5.2 Confusion Matrix

Random Forest model had mean ROC-AUC of 0.830 (±0.028), which was a bit lower than the Logistic Regression. Although this model had a good performance in terms of distinguishing non-upgrade customers (recall = 0.92), the recall on upgrade customers was moderate (0.49).

Random Forest was in the form of an ensemble and thus was able to learn non-linear interactions but this did not lead to better performance on the upgrade task. The general accuracy was 81% which means good but not best performance than the Logistic Regression.

5.1.3 XGBoost Results (Upgrade Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.8200**
Standard Deviation (ROC–AUC)	**0.0279**

Table 5.5 ROC-AUC Performance

Fig. 5.3 Confusion Matrix

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.85	0.87	0.86	719
Yes	0.59	0.53	0.56	244
Accuracy	—	—	**0.79**	963
Macro Average	0.72	0.70	0.71	963
Weighted Average	0.78	0.79	0.78	963

Table 5.6 Classification Report

XGBoost has an average ROC-AUC of 0.820 (±0.028). The upgrade class had a balanced precision and a recall value of 0.59 and 0.53 respectively. Although it is capable of modeling complex patterns, XGBoost failed to do so better than the simpler models in this task.

Comparative Insight:

Logistic Regression was always the most effective between the three models, which points to the fact that relatively structured and linear signals of engagement influence upgrade behavior.

5.2 Performance Evaluation of Cross-Sell Prediction Models

The cross-sell prediction tries to establish the customers to buy more products or services. This task was more complex and varied than the one of prediction of upgrades.

5.2.1 Logistic Regression Results (Cross-Sell Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.4569**
Standard Deviation (ROC–AUC)	**0.0235**

Table 5.7 ROC-AUC Performance

Fig. 5.4 Confusion Matrix

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.26	0.02	0.04	376
Yes	0.60	0.96	0.74	587
Accuracy	—	—	**0.59**	963
Macro Average	0.43	0.49	0.39	963
Weighted Average	0.47	0.59	0.47	963

Table 5.8 Classification Report

Logistic Regression was not a useful predictor of cross-sell with a mean ROC-AUC of 0.457 (±0.024). The confusion matrix indicates that there is a high bias in predicting the positive class,

hence high recall of 0.96 on cross-sell customers, and an exceedingly high recall (0.02) on non-cross-sell customers.

This imbalance resulted in a humble overall accuracy of 59%, which means that though the model is aggressive in identifying the potential cross-sell customers it is not sufficiently discriminatory.

Interpretation:

It seems that cross-selling behavior is a non-linear phenomenon with heterogeneous factors that cannot be effectively modeled with the help of a linear classifier.

5.2.2 Random Forest Results (Cross-Sell Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.5287**
Standard Deviation (ROC–AUC)	**0.0503**

Table 5.9 ROC-AUC Performance

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.41	0.20	0.27	376
Yes	0.61	0.82	0.70	587
Accuracy	—	—	**0.58**	963
Macro Average	0.51	0.51	0.49	963
Weighted Average	0.54	0.58	0.53	963

Table 5.10 Classification Report

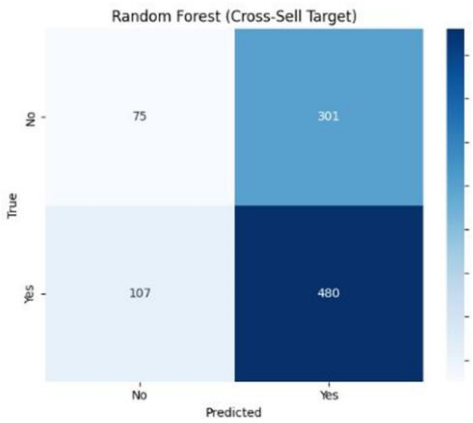

Fig. 5.5 Confusion Matrix

Random Forest model had a better performance in terms of mean ROC-AUC of 0.529 (±0.050). It had better balance of precision and recall of both classes as compared to Logistic Regression. The model rightly determined more non-cross-sell customers and importantly, good recall (0.82) of cross-sell customers.

This notwithstanding, the level of misclassification was high, which is attributable to the complexity of cross-selling behavior.

5.2.3 XGBoost Results (Cross-Sell Target)

ROC–AUC Performance

Metric	Value
Mean ROC–AUC	**0.4672**
Standard Deviation (ROC–AUC)	**0.0359**

Table 5.11 ROC-AUC Performance

Classification Report

Class	Precision	Recall	F1-Score	Support
No	0.35	0.28	0.32	376
Yes	0.59	0.67	0.63	587
Accuracy	—	—	**0.52**	963
Macro Average	0.47	0.48	0.47	963
Weighted Average	0.50	0.52	0.51	963

Table 5.12 Classification Report

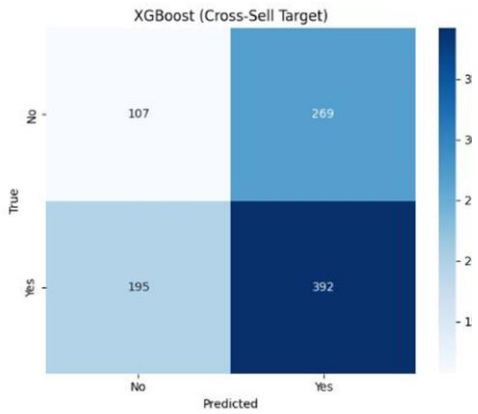

Fig. 5.6 Confusion Matrix

The mean ROC-AUC obtained by XGBoost was 0.467 (±0.036), where the balance between classes was better than that of Logistic Regression. The model showed moderate recall (0.67) and precision (0.59) of cross sell customers with a total accuracy of 52%.

Comparative Insight:

Ensemble models based on trees were found to be more adaptable to cross-sell prediction because they could model intricate patterns of engagement, but had inferior predictive performance in general as compared to upgrade models.

5.3 Comparative Model Analysis

An experimental comparison of predictive models corresponding to upgrade and cross-sell targets indicates significant variance in performance due to inherent complexities in behavior of every task. In the case of upgrade prediction, Logistic Regression was always able to perform better than the ensemble models, with higher and more stable values of ROC-AUC. It means that the structured and linear patterns of engagement play a major role in the process of making upgrade decisions, and it can be well represented through the simpler parametric models. Conversely, cross-sell prediction demonstrated a poorer overall predictive performance compared to all the models indicating that the cross-sell prediction has more behavioral uncertainty. Nevertheless, non-linear ensemble methods including Random Forest, XGBoost showed rather improved results compared to the Logistic Regression in this task, which indicates

that the cross-selling behavior is related to the involvement of more sophisticated feature interactions. On the whole, the findings approve that the predictable behavior of upgrades is much higher than the predictable behavior of cross-sells, which shows that it is essential to match the complexity of the model with the specifics of the prediction issue.

5.4 Business Interpretation of Results

The findings provide various practical implications to subscription-based service providers in the view of management and strategies. The high accuracy of predicting the upgrade is why the upgrade prediction model allows the firms to actively detect customers who are highly likely to switch to premium plans, which makes it easy to allocate the marketing resources and promotional campaigns to the correct customers. By comparison, cross-sell prediction models are less accurate, but they still can be useful as decision support tools in combination with domain knowledge and business rules, especially when it comes to personalized content or add-on recommendations. The findings also indicate that the choice of the model to be used should be based on the structure of the problem instead of the sophistication of the algorithm because a simple model can be more effective than the sophisticated algorithms as long as the customer behavior can be described by simple and explainable patterns. This understanding can be utilized to assist organizations to strike a balance between predictive accuracy and interpretability and operational feasibility.

5.5 Key Findings

The research has a number of major findings that can be applied to both theory and practice. First, intent-based target construction was used and substantially increased the balance of the classes and model stability, which is problematic in imbalanced subscription datasets. Second, upgrading behavior was observed to have stronger and consistent signals of engagement than cross-selling behavior, hence it exhibited the highest predictive accuracy in all the models. Third, the findings show that a simple model like Logistic Regression may be better than complicated ensemble algorithms in the case when the process of customer decisions has a structure and is linear. Lastly, cross-selling is also a predictive problem that is difficult, which shows that more behavioral, contextual, or temporal variables are required to enhance model performance in future studies.

6. Discussion

In this research, there was a comparison between machine learning models in the prediction of subscription upgrades and cross-selling behavior. The results indicate that there are evident disparities between the two prediction tasks. The Upgrade prediction was significantly better in all models, and the Logistic Regression provided the highest results. This implies that the patterns of engagement which are relatively linear that include usage frequency and subscription tenure are used to drive upgrade decisions. Customers with a gradual increase in their use of services are likely to upgrade in the predictable way.

Conversely, the behavior of cross-selling was much harder to predict. The models yielded all lower values of ROC-AUC, which is a sign of weak discriminatory strength. This means that the decision of cross-sell depends on more complicated and situational factors than simple engagement measures. Although the tree-based models slightly enhanced the performance, the improvements were minimal, and it could be believed that more comprehensive or more diverse features can be needed to reflect the cross-sell intent better.

The other valuable observation is that the complexity of a model does not necessarily result in high performance. The successful performance of Logistic Regression on the upgrade prediction indicates that simplified and explainable models would be more effective than sophisticated ensemble tools in the case of the upgrade situation when the customer behavior exhibits regular trends. In general, the research points out the role of matching the model choice and the behavioral structure, and it is necessary to note that the various customer conversion goals need varied predictive approaches.

7. Implications of the Study

7.1 Theoretical and Academic Implications

The present study adds to the body of academic literature on behavioral analytics, subscription-based business models and data-driven marketing since it empirically shows that machine learning methods are effective in predicting customer monetization opportunities using engagement data. The results are an extension of the weight of current theoretical expectations because they confirm the role of behavioral intensity (including viewing hours, content-consumption, and watchlist size), and feature-access limits as a primary factor that predicts potential cross-selling and upgrading. The effective predictive abilities of cross-selling performance are in line with the content of usage-based and engagement-driven consumer behavior hypotheses, that with increased consumption, the perceived value and readiness to complementary products rises. Besides, the excellent functionality of ensemble models underscores the existence of non-linear and interaction-based behavioral mechanisms, which supports the limitation of the traditional linear models in describing complex customer decision-making procedures.

By contrast, the moderate predictive power of subscription upgrade behavior indicates that there is a significant theoretical difference between incremental monetization and long-term commitment escalation. Although the decisions which are made on cross-selling may be based on observable patterns of engagement, upgrade decisions seem to have latent psychological, economic, and contextual issues including sensitivity to price, perceived switching costs, and customer loyalty. This difference adds value to theory by implying that subscription upgrades are a more complex order of decision-making process that cannot be well explained by usage intensity over the short term. Also, the paper develops methodology theory through the combination of business-rule-based target creation and supervised machine learning, providing a reproducible, interpretable framework to the connection between conceptual constructs and operational analytics in online subscription systems.

7.2 Suggestions for actions or strategies based on research findings

Relying on the results of the empirical analysis and the predictive data of the existence of the machine learning models, it is possible to suggest a range of effective steps towards improving revenue optimization and customer value management in subscription-based digital platforms. The above high predictive accuracy of the cross-selling means that organizations should focus on data-based cross-selling models by focusing on high-commitment customers who possess unexploited feature needs, including the absence of multi-device access or parental controls. Enhancing the precision of targeting in real-time recommendation engines by incorporating ensemble models such as XGBoost, or Random Forest, can lead to substantial improvements in marketing inefficiency, targeting, and customer fatigue such as that experienced by consumers of generic promotion campaigns. Also, by incorporating behavioral segmentation into marketing automation workflows, firms can customize communication, depending on their use intensity, content consumption behavior and watchlist behavior, and they can therefore better match their promotional message to the customer-preferred choice.

Conversely, the comparatively moderate results of the upgrade prediction models indicate that subscription upgrades need to take a more multi-dimensional and holistic approach. Organizations need to add behavioral analytics with contextual variables, including customer loyalty, loyalty incentives, the ability to flex prices, and promotional offers that have time constraints to improve the impact of upgrading decisions. Predictive accuracy can be further improved in dynamic markets by continuous retraining of the model and adding features (temporal use patterns, churn features, and customer feedback). Lastly, the implementation of artificial intelligence-based decision support systems, which incorporate predictive insights on strategic dashboard can equip decision-makers to act proactively to manage customer relationships, maximize monetization pathways, and maximize customer lifetime value.

7.3 Practical implications for stakeholders or decision-makers

The results of the research offer valuable practical implications to the stakeholders in marketing, product management, analytics, and strategic decision-making in the field of subscription-based digital platforms. The high predictive accuracy of machine learning models on cross-selling

helps organizations to use data-driven and personalized approaches to engagement moving away to targeting based on intuition. By using these insights, marketing managers can develop targeted campaigns that advertise the add-on features like multi-device access or premium content packages, which will enhance the effectiveness of the campaigns, save marketing resources, and boosting the conversion rates as the marketing team will reach the target customers with a specific set of targeting.

To product and subscription managers, the average capability of the upgrade prediction models shows that subscription upgrades need a more delicate strategy than the behavioral engagement measures. Value-based communication, flexible pricing schemes, features trial, and loyalty should be encouraged as upgrade strategies to meet the different levels of customer preparedness. Operationally, the incorporation of predictive outputs into decision-support dashboards can be utilized to enable the leadership teams to proactively identify the monetization opportunities and maximize customer lifetime value. On the whole, the implementation of these insights may help to ensure the growth of revenues, better customer satisfaction, and more competitive positioning in data-driven subscription markets.

8. Conclusion

8.1 Summary of key points discussed in the project

The current project explored how machine learning-based behavioral analytics could be used to forecast potential to upgrade subscription and cross-selling in a digital service setting that is subscription-based. The research design adopted in the study was based on a quantitative, predictive, and exploratory design that used a publicly available dataset to examine the customer engagement patterns and monetization readiness. Target variable construction was done using business-driven logic to make sure that the target variables were practical and interpretable. Exploratory data analysis showed that there were strong relationships between engagement measures including viewing hours, watchlist size, and content downloads and cross-selling potential. The performance of the ensemble machine learning models was best observed with cross-selling prediction, especially the XGBoost and the Random Forest, whereas, upgrade prediction was of an intermediate level of accuracy, which underscored the complexity of decisions made on long-term commitment. The strength and dependability of the analytical framework was confirmed by model assessment through the use of confusion matrices, classification reports, and ROC-AUC curves.

8.2 Reflection on the project process and outcomes

The project took a logical and consecutive analytical procedure starting with exploration of data, moving to feature engineering, building targets, and predictive modeling. The fact that the study combined business logic with machine learning is one of the major strengths of the study since the analytical results were not only statistically valid but also strategic. The findings proved that behavioral engagement data can be successfully used to support strategies to optimize revenues, especially, cross-selling. Nevertheless, the relative low score in upgrade prediction highlights the weaknesses of using observable engagement measures alone. This reflection emphasises the need to balance analytical rigour and contextual business perception in order to obtain actionable insights.

8.3 Limitations of the Study

Although the findings in this study have value to offer to the study of behavioral analytics in the context of subscription upgrades and cross-selling, there are a few limitations that must be taken into consideration when making meaning out of the results. First, the analysis is performed on the basis of one secondary data received in a public source, which can limit the possibility to generalize the findings to other subscription-based industries, models, and customer demographics. Second, the research is not mostly based on the observable behavioral and engagement variables hence ruling out latent factors like price sensitivity, promotional exposure, attitudinal loyalty, and psychological motivations which have a major role in making an upgrade decision as indicated by a moderate predictive performance of upgrade models. Third, rule-based business logic constructions of target variables, though increasing interpretability, add subjectivity and fail to verify cause and effect relationships between predictors and outcomes. Also, the train-test split is not random, and that is why the temporal stability cannot be fully evaluated since the customer behavior can vary depending on the time. Lastly, predictive framework is concerned with the classification accuracy and not causal validation, that is, the identified high-propensity customers will not always translate into a real-life situation of intervention. Such limitations point to the opportunities of future research to use more appropriate data sources, longitudinal validation, and causal modeling techniques.

8.4 Recommendations for future research or areas of further study

Future studies could help them improve the predictive performance through the inclusion of longitudinal and time-based data, including tenure of the customer, how they used those promotions over time, and how they reacted to the past promotions. Pricing sensitivity indicators, churn history, and customer feedback data should be included in the model because it would enhance the prediction of upward behavior. Furthermore, a more sophisticated approach like deep learning, a survival model, or reinforcement learning can provide more valuable information about customer lifecycle changes. The results could be further generalized by comparing them to other industries, including OTT services, SaaS offerings, or EdTech subscriptions. The investigations into the causal inference can also assist in making a clear differentiation between correlation and causation in the customer engagement patterns.

8.5 Concluding Remarks

To sum up, this paper has shown that machine learning-based behavioral analytics has the strategic importance of improving decision-making in business models that are based on subscriptions. The research will be useful both to the academic and the real-world field of data-driven marketing strategy as it has successfully identified customers who can be cross-sold and provided information about the upgrade behavior. The results support the necessity to use advanced analytics to create personal customer engagement, sustainable revenue growth, and competitive advantage in the digital markets. All in all, the project provides a great basis on the new analytics-based innovations related to subscription monetization and customer behavior modeling in the future.

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